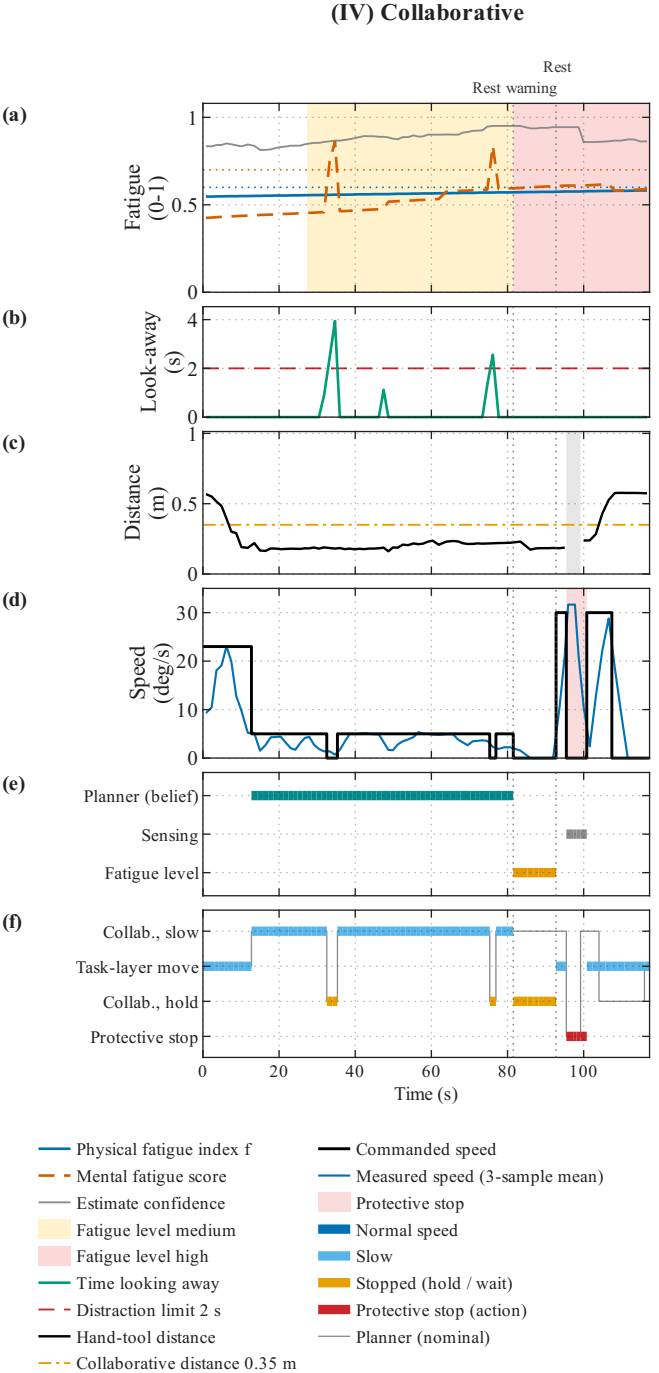

(IV) Collaborative
Rest
Rest warning
(a)
Fatigue (0-1)
1
0.5
0
(b)
Look-away (s)
4
2
0
(c)
Distance (m)
1
0.5
0
(d)
Speed (deg/s)
30
20
10
0
(e)
Planner (belief)
Sensing
Fatigue level
(f)
Collab., slow
Task-layer move
Collab., hold
Protective stop
0
20
40
60
80
100
Time (s)
Physical fatigue index f
Mental fatigue score
Estimate confidence
Fatigue level medium
Fatigue level high
Time looking away
Distraction limit 2 s
Hand-tool distance
Collaborative distance 0.35 m
Commanded speed
Measured speed (3-sample mean)
Protective stop
Normal speed
Slow
Stopped (hold / wait)
Protective stop (action)
Planner (nominal)